Algebraic manipulation

1

a $x = 5, -5$ **b** $x = \pm 7\sqrt{2}$ **c** $x = \frac{8}{9}, \frac{-8}{9}$ **d** $p = \pm 12$

e $x = 0, \frac{12}{7}$ **f** $x = 4 \pm \sqrt{10}$ **g** $x = \frac{5}{4}, -\frac{11}{4}$ **h** $x = \frac{9}{4}$

i $x = 1$ **j** $m = 0, 10$

2 $x = \pm\sqrt{\frac{c}{a}}$

Factorisation

3

a $x = 0, -12$ **b** $y = 0, \frac{1}{5}$ **c** $m = 0, 14$ **d** $x = -10, -9$

e $m = 5, -4$ **f** $x = 10$ **g** $x = -4, 5$ **h** $m = 14, 13$

i $x = 8, -3$ **j** $y = -1, 3$ **k** $x = 2, -4$ **l** $x = \frac{9}{4}, 4$

m $x = -\frac{5}{4}, -3$ **n** $x = -\frac{4}{5}, -2$ **o** $x = \frac{5}{4}, 3$ **p** $x = -\frac{5}{3}, 4$

q $x = \frac{-5}{2}, 12$ **r** $b = -\frac{5}{3}, \frac{3}{4}$ **s** $k = 1, \frac{7}{6}$

4 $x = \pm 4$

5 $x = 2$

6

a $(x - 7)(x + 4) = 0$ **b** $x(x + 1) = 0$

7 $x^2 + 8x + 15 = 0$

Using technology

8

a $x = 2, -\frac{5}{3}$ **b** $x = \frac{3}{2}, -\frac{7}{2}$ **c** $x = 0.4, -2.0$

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9

a $x = \pm\sqrt{2}$

b $x = 1.4, -1.4$

10

a 1

b 1

Applications

11 None

12 $x = \dfrac{12 \pm \sqrt{1044}}{90}$

13 $x = -\dfrac{3}{5}, -2, -\dfrac{5}{2}, -4$

14 $k = -16$

15 $a = 13$

16 $t = 22, 14$

17

a $t = 4, 8$

b 4 hours

c 4 hours

18

a $x = 0, 32$

b 32 cm

19

a $17 + \dfrac{6426}{x^2} = \dfrac{663}{x}$

b $x = 18, 21$

c 36.83 cm

20

a $0, 14$ months

b No, after one year the company will still be spending more money than it is making.

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